Documentation Portfolio

Engineering Design | The AzotoColumn

2840 - 2

Nashville, Tennessee 2025

Table of Contents

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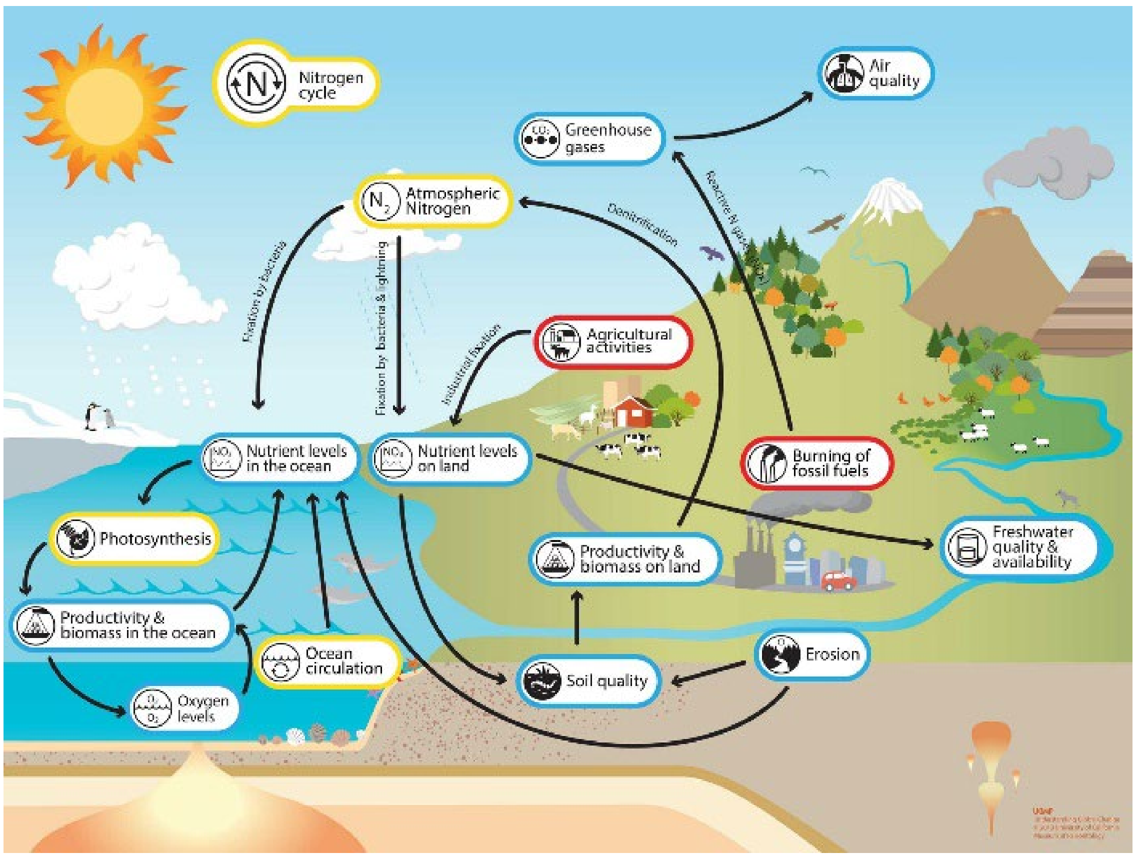
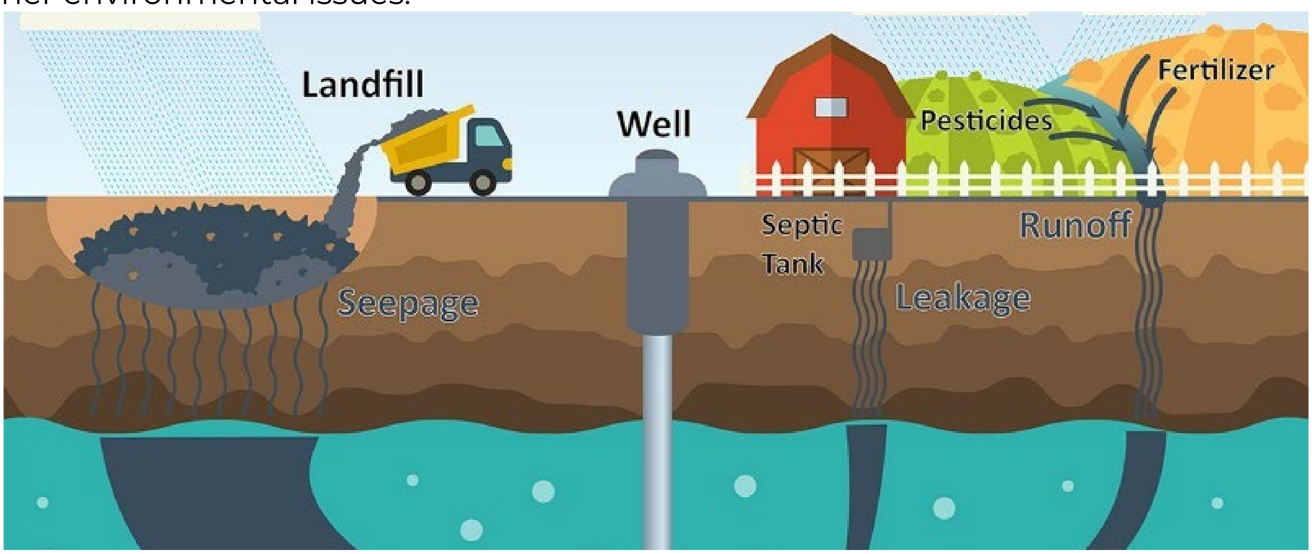
Imagine a world where the very air we breathe and the food we grow turn against us due to an imbalanced nitrogen cycle—yet this critical process, one of the key parameters that enable Earth to enjoy its relative stability. It poses problems, though, which render it an inconvenience to the world. Nitrogen oxides, when released into the atmosphere, could inadvertently spawn acid rain should it become excessive, while nitrogen-based heavy fertilizers in the soil promote leaching which ultimately led to harmful runoffs. Finally, plants without the lack of denitrifying bacteria will also not grow as well. The nitrogen cycle regulates the number of forms of nitrogen on the planet, but only as much as it destroys it.



The continuous emissions of nitrogen oxides owing to transport, combustion of fossil fuel, industrial process, and agricultural practices results in air pollution and smog formation. These nitrogen oxides react with atmospheric water vapor to form nitric acid, which falls as acid rain. Acid rain lowers the pH of water bodies and soils, which damages aquatic life, soil fertility, and vegetation, causing loss of biodiversity. Moreover, excessive application of nitrogen-based fertilizers in agricultural industries leads to an overload of nitrogen leaking into water bodies through a process known as eutrophication. Nitrogen, in the form of nitrates, especially stimulates the overgrowth of algae in water. This may lead to the formation of "algal blooms," which die and decay in water, consuming oxygen and thus forming "dead zones" in which aquatic organisms cannot survive. It disrupts marine food chains, depletes fisheries, and reduces fisheries' productivity.

Finally, fertilizer nitrogen in runoff can pollute the groundwater, the source of drinking water. High levels of nitrates in drinking water, common in rural areas as partially a result of agricultural runoff, have become a human health concern because these disrupt the metabolism of oxygen in humans, especially a condition in babies known as blue baby syndrome because of its effect on the transport of oxygen in hemoglobin. It can also destroy terrestrial ecosystems by disrupting the normal nitrogen balance in the soil and causing them to eventually become infertile. Managing the nitrogen cycle is important due to the key role it plays in ecosystems, agriculture, and the environment, all of which have a large impact on humans. The uniqueness of the nitrogen cycle stems from nitrogen gas (N2) being primarily unusable by most organisms in its natural state. To make it accessible, it must go through fixation, converting it to forms like ammonia or nitrate, which are usable for plants. Humans have had a major role in disrupting the nitrogen cycle, with the use of rich nitrogen fertilizers and fossil fuels causing eutrophication and other environmental issues.



Over time, farmers have become more aware about the amount of nitrogen emitted due to fertilizer. Excess nitrogen from fertilizers can lead to leaching and runoff, including detrimental side effects like soil degradation. Not only does nitrogen affect the soil, but nitrous oxide, a greenhouse gas, can be released into the atmosphere, thereby contributing to the rise in global temperature. As a result, many farmers have adopted sustainable practices, such as precision agriculture, to minimize nitrogen runoff. Moreover, technological advancements including soil testing and sensor-based applications help farmers stop overapplying nitrogen heavy fertilizers. Regulations enforced by the government, such as a tax on nitrogen emissions, have contributed to the prominence of these eco-friendly practices. By managing nitrogen emissions much more effectively, farmers can improve soil health, protect water sources, and contribute to mitigating climate change.

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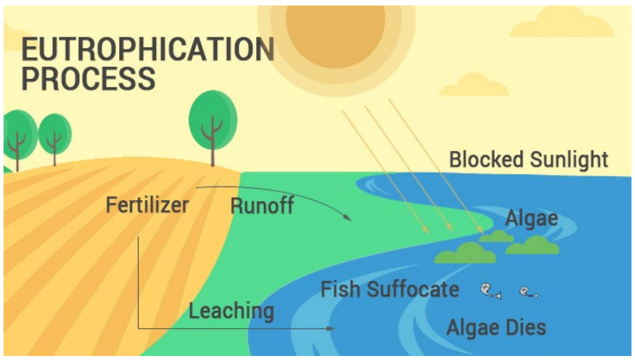

technological advancements including soil testing and sensor-based applications help farmers stop overapplying nitrogen heavy fertilizers. Regulations enforced by the government, such as a tax on nitrogen emissions, have contributed to the prominence of these eco-friendly practices. By managing nitrogen emissions much more effectively, farmers can improve soil health, protect water sources, and contribute to mitigating climate change.

Farmer Awareness and Sustainable Practices

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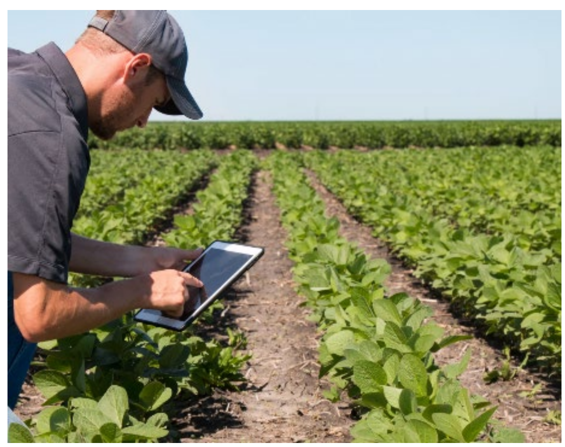


Finding Solutions and Scientific Breakthroughs

Finding solutions to properly regulate the nitrogen cycle will reduce environmental damage and improve food security, as nitrogen is essential for producing enough plants to sustain human population growth.

In the past, there have been several major scientific breakthroughs such as the Haber-Bosch process. The Haber-Bosch process artificially converts hydrogen and nitrogen to ammonia and supports plant growth due to nitrogen fixing bacteria being a limiting factor of fixation. In our APES class this year, we learned about how the nitrogen cycle affects all layers of an eco-column (terrestrial, decomposition, aquatic), and the flow of nitrogen through each layer. The decomposers, including nitrogen fixing bacteria, fix nitrogen from the atmosphere and introduce it to the layers. Plants and animals in other layers will then absorb it for their respective uses, and it will be released back into the atmosphere by denitrifying bacteria. An eco column is nothing but a simulation of a real ecosystem, so with these findings in mind, we decided to have a healthy mix of these types of bacteria in most of our potential solutions in order to maintain a balanced nitrogen cycle.

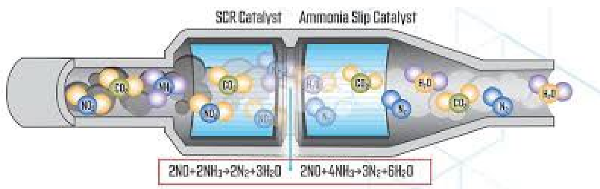
Research for Tech in Solution: Precision Agriculture

Precision agriculture

- Utilized advanced agricultural technology such as GPS guided machinery and soil sensors
- Optimizes fertilizer application across a certain culture of crops.
- Results show that farmers who employ precision agriculture have cut fertilizer usage by 30%, thereby minimizing runoff

Tech Inside Solution Connections: Biological Nitrogen Fixation

- Biological nitrogen fixation allows plants to absorb atmospheric nitrogen.
- Non-leguminous plants (those without root nodules or nitrogen-fixing bacteria) can benefit too.
- This is done using seed-inoculated bacteria (bacteria applied to seeds at the time of sowing).
- As a result, the need for synthetic fertilizer can be reduced by up to 50%.

Tech Inside Solution Connections: Selective Catalytic Reduction (SCR) 	• Nitrogen can get emitted from vehicles and factories. • One way to mitigate this issue is using selective catalytic reduction (SCR). • This process converts nitrogen oxides into harmless nitrogen and water using catalysis, which achieves an 80% reduction of nitrogen emissions in power plants. • Usually, these methods will be implemented in factories, and in the future, even in vehicles.
Tech Inside Solution Connections: Cleaner Fuel **Protocol** 	A simpler solution could be to adopt a cleaner fuel protocol, meaning the rise of electric vehicles or more efficient MPG rates. Ultimately, this method will reduce transportation related nitrogen oxide emissions, leading to less excess nitrogen in the atmosphere and environments in general.

Summary of Sources:

- Source 1 Nature Education: In this article, the nitrogen cycle is discussed regarding its important processes, like nitrogen fixation, nitrification, and denitrification and the organisms that play a role in it.
- Source 2 Susdrain: This source describes bioretention basins as shallow, vegetated depressions for capturing, filtering, and treating stormwater runoff. This chapter is concerned with the implementation of SuDS, comprising treatment of the captured water from rainfall events and green infrastructures.
- Source 3: MIT News: This source alludes to genetic engineering of crops like wheat and rice for self-fixation of atmospheric nitrogen without the intervention of synthetic fertilizers. This will increase food security and make farming more viable for parts of the world where fertilizers are not available.
- Source 4 Science Direct: The article is an overview of selective catalytic reduction (SCR), a process for the reduction of nitrogen oxide (NOx) emissions from vehicle and industrial exhaust. It is a summary of SCR's effectiveness, mechanisms, and advantages, which render it a critical emissions control strategy in environmental management.
- Source 5 Nitrogen Control: This article narrates that nitrogen is a gas that is very common in the atmosphere and explains that too much nitrogen in the atmosphere

can start to deteriorate air quality and cause acid precipitation. On the contrary, improvements can be made by simply reducing nitrogen emissions by using methods such as taxing nitrogen emissions.

- Source 6 Eutrophication: One of the many problems of excess nitrogen emissions is eutrophication, where bodies of water become dead zones due to lack of oxygen. There are 6 main steps to this process: nitrogen leaching, algae blooms, algae die, bacteria decompose matter, biological oxygen demand, and then dead zone created.
- Source 7 Eco Column: This source explains what an eco-column is and how energy is cycled throughout each of the three layers of the simulated environment.

Sources listed in resources.

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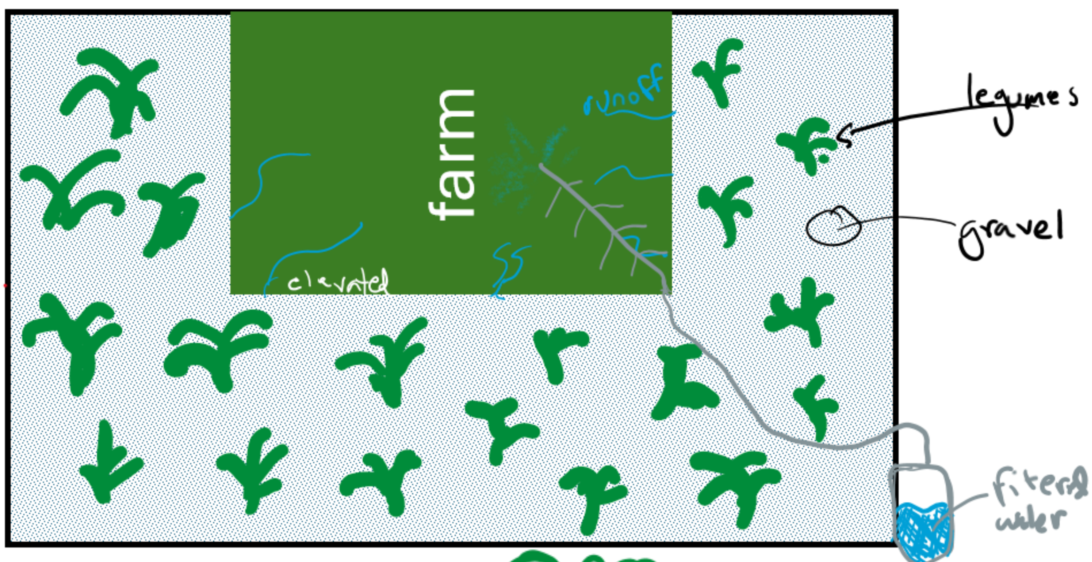
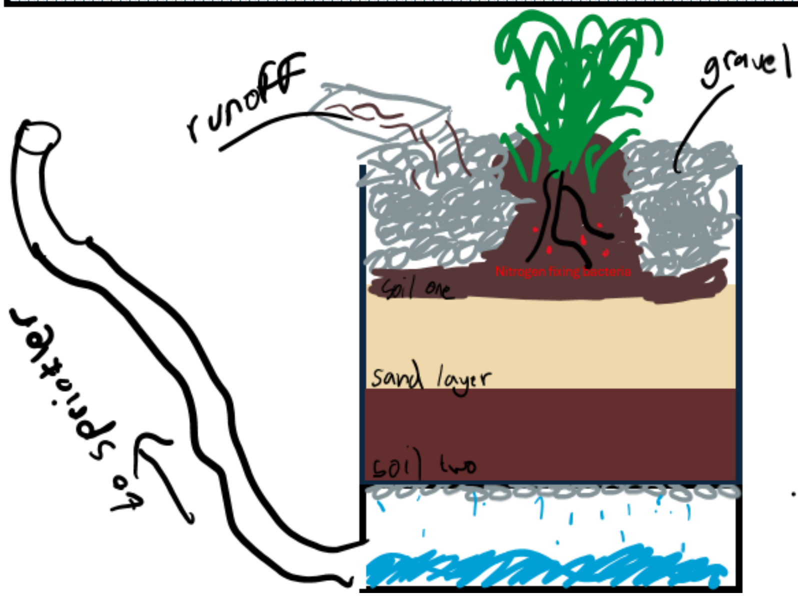
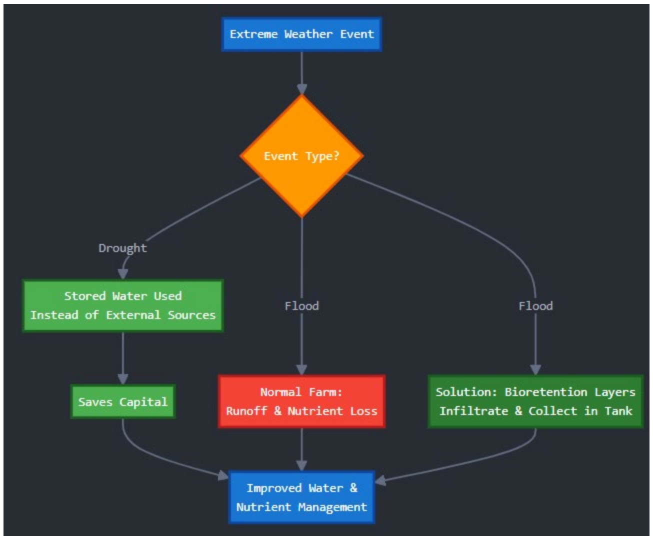
Our first solution uses a specifically designed bioretention area that collects runoff from farms, which are usually littered with harmful fertilizers. This Bioretention area is a 10 feet radius around the farm. It is also 2 feet below it so that gravity carries the runoff directly to it. The fertilizer is mixed with runoff water that is purified through many intricate steps spanning 6 feet underground in the retention area and is then pumped back up where it is redistributed to the plants via sprinklers for consistent irrigation. The recovered fertilizer is placed back in soil, sustaining plant growth while reducing waste. This system is water-saving, reduces the need for additional fertilizer, and provides a way to do agriculture that is sustainable and eco-friendly.

Initial Idealization Sketches:





Flowchart for water flow during droughts and floods with bioretention:



Additionally, the water retention tank can be beneficial during extreme weather events, such as droughts or floods. In the case of a drought, the excess water stored in the tanks can be used instead of utilizing water from external sources, which saves a significant amount of capital. On the other hand, during a flood, water would runoff in greater amounts on a normal farm, resulting in increased leaching and runoff of nutrients. With the proposed solution, most flood water will be infiltrated through the many layers of the bioretention area, and the excess water will be collected by the tank; both processes are driven by gravitational pull.

Steps:

1. Excess water from the high farm or the road enters the bioretention area through the flow of gravity.
2. It carries particulates, biohazards, and nutrients that need to be filtered before reuse.
3. Water flows into border areas around the farms that are vegetated with legumes, including alfalfa, red clover, crimson clover, hairy vetch, pigeon pea, and peas.
4. These plants, in symbiosis with soil bacteria, absorb nitrates and improve soil health.
5. Plants grow in soil surrounded by gravel, which aids in the filtration process.
6. Gravel zone: removes the large debris and sediment.
7. Sand zone: filters smaller particles for additional purification steps.
8. Soil zone: absorbs pollutants, removes nitrates, and improves bacterial denitrification, hence improving the nutrient uptake of legumes.
9. Filtered water is collected at the base of the bioretention area.
10. A solar-powered pump located at one corner of the farm is turned on when the base is full and pumps the water out.
11. The clean filtered water is pumped or gravity-fed back to the farm for irrigation.

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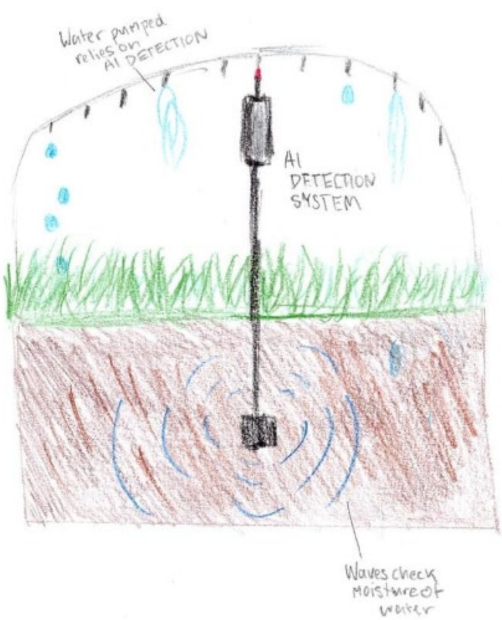
Our second solution is to use an AI-based system to monitor and detect nitrogen levels and the presence of denitrifying bacteria in the soil. It uses sensors placed on stakes across farmland to monitor and analyze individual sectors of land. These sensors determine the exact concentration of fertilizer needed using a machine learning model, ensuring precision in resource use. Additionally, the sensors must have scalability to be deployed en masse and field durability to accommodate extreme weather events. This is important because nitrogen in the air (N_2) is not directly usable by plants. Denitrifying bacteria are essential for converting nitrogen into forms plants can use, such as nitrates (NO_3^-) and ammonium (NH_4^+). The system identifies areas where bacteria concentration is low and applies natural fertilizer to support their activity, enhancing the natural nitrogen cycle. Moreover, data synthesis of sensor and geospatial variables helps the model integrate the sensor data and adapt it to conditions in the outside world. Choosing the most efficient algorithm determines how accurate and effective the model will be. Most importantly, calibration requirements must be set in order to confirm the reliability and future success of the system, and the training helps with conditions the system may face on the farm. Using these technologies will increase the effectiveness of mitigating nitrogen runoff.

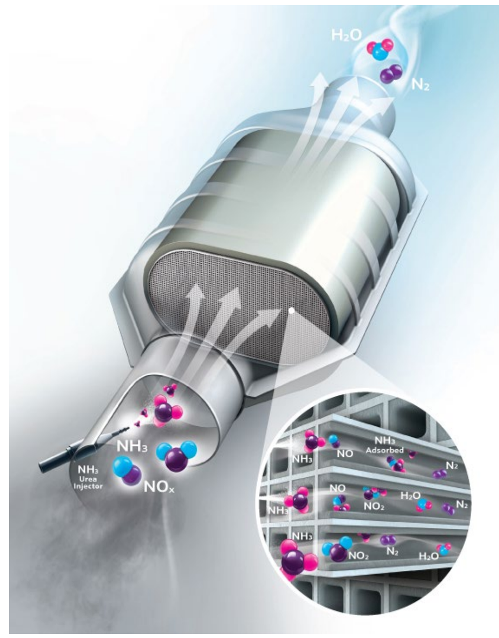


Benefits of Solution 2:

1. **Optimizing Resources:** The sensors monitor nitrogen levels and detect the presence of denitrifying bacteria using molecular techniques like quantitative PCR (qPCR). By targeting specific genes within the denitrification pathway, the system ensures water and fertilizer are applied only when and where they're needed. This reduces waste, lowers irrigation costs, and saves reso urces.
2. **Reducing Nutrient Runoff:** Overusing fertilizer can lead to runoff which pollutes nearby water bodies and causes issues like eutrophication. Our system prevents this by applying only the required amount of fertilizer, protecting ecosystems and promoting sustainable farming practices.

Solution #3

Our third solution is to use selective catalytic reduction technology to reduce nitrogen emissions in forms of transportation that require fuel to be burnt. This works by injecting a reducing agent into the exhaust gas, which produces ammonia when vaporized. The ammonia reacts with NOx in the presence of a catalyst, creating water and nitrogen gas. Nitrogen gas isn't as harmful as NOx when emitted, as NOx creates acid rain.



SCR technology is also used in different industries, such as transportation and factories, to reduce harmful emissions. It's a practical solution that works in many types of machines and vehicles. The system is designed to be durable and cost-effective when properly maintained. By cutting down on NOx emissions, SCR helps improve air quality and reduces the damage caused to the environment.

Benefits of selective catalytic reduction (SCR) technology:

1. Environmental Impact: It reduces nitric oxide emissions by converting them into nitrogen gas and water, decreases acid rain to protect aquatic ecosystems and soil quality, and improves air quality by minimizing smog.
2. Proven Effectiveness: SCR is a reliable technology already used in heavy-duty diesel engines and industrial sectors, and it can be integrated into existing vehicle infrastructures.

Besides vehicles, SCR systems are important to cut down emissions across major industries, which include power generation, mining, and rail transport (trains). These systems are most effective at higher temperatures and rely on parts like

diesel exhaust fluid tanks to do their job. While there is a small amount of CO_2 produced during the process, it's negligible compared to environmental benefits, especially when SCR can cut NO_x emissions by up to 90%. With its proven effectiveness and high adaptability, SCR isn't just a helpful tool; it could prove to be a vital part of our future.

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We made a detailed pros and cons table along with a Pugh chart in order to compare our prototypes in order to select the most feasible and effective solution. These tables are listed below.

Table 1 - Pros and Cons

Design	Pros	Cons
Bioretention Area	- Reduces nitrogen runoff in bodies of water. - Enhances soil and plant health. - Requires minimal maintenance once established.	- Takes up a lot of space, so not great for small farms. - Setting it up is a pain and it is a lot of digging and layering. - We must pick the right soil or it won't work as well.
AI Monitoring System	- Saves time and makes the whole process more efficient. - Gives real-time data to help with fertilizer use.	- Expensive upfront, especially for sensors and setting up the system. - Sensors need maintenance, so it's not totally hands-off. - Some people might not trust AI taking over farm decisions.
Catalytic Reduction	- Reduces nitrogen emissions into the atmosphere	- Needs specialized equipment and trained people to use it. - Only works under certain conditions, so it's not a one-size-fits-all fix. - Uses a lot of energy, which kinda defeats the whole "helping the environment" idea..

Table 2 - Pugh Chart
Key: **+** indicates positive, **0** indicates neutral, **-** indicates negative

Criteria	Bioretention Area	AI Monitoring System	Catalytic Reduction
Cost	-	-	-
Water Resistance	+	+	0
Size	-	0	+
Effectiveness	+	+	+
Accessibility	+	-	-
Maintenance	-	-	-
Waste Generation	+	+	0
Response Time	0	+	-
Safety	+	0	-
Total	+2	0	-2
Comparative Ranking	1st	2nd	3rd

Using these tables combined with evidence gathered from online research, we came to the conclusion that the best solution is the Bioretention Area. We decided that this solution would be best modeled through an initial diagram, shown next. Around this time we also named our solution the AzotoColumn, “Azoto” meaning nitrogen in Greek.

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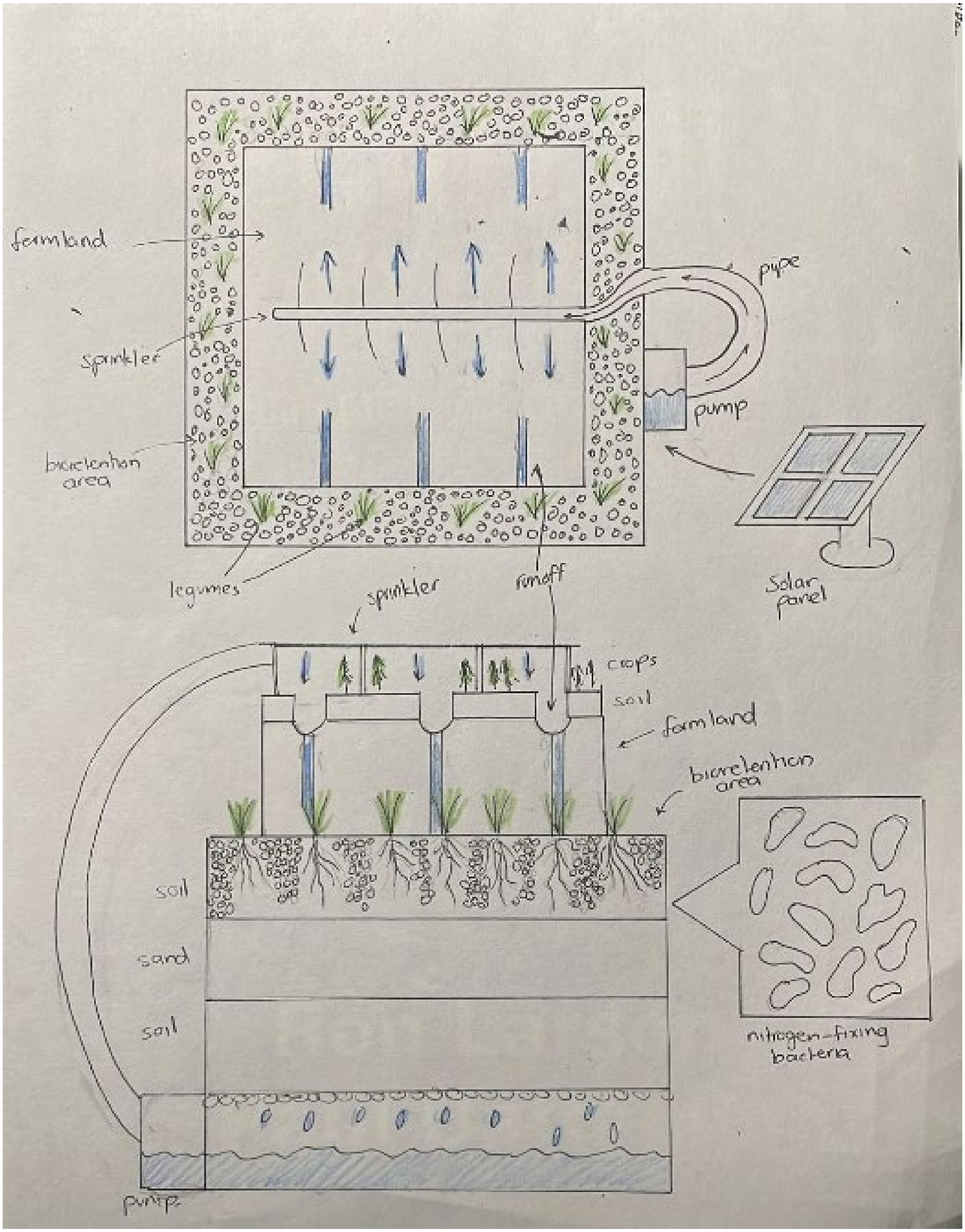


I. Means of testing Solution

DISCLAIMER: OUR SOLUTION CAN BE TESTED AND WAS; THESE DETAILS HELP THE READER UNDERSTAND OUR MEANS OF GOING ABOUT THAT BETTER.

To evaluate our solution, two main areas were tested: the efficiency of water movement and the nitrogen levels in both the soil and the blocked-off runoff. Prior to evaluating our solution, we wanted to calculate soil composition in order to check our data and more accurately compare it with real life systems. Below is the procedure for finding our soil composition:

Soil Tests

Objective: Determine soil composition and pH for use in all bioretention system tests.

Procedure:

1. Examine the sample for plant material, minerals, and color. Dark soil likely has more organic material, while light soil is more mineral based.
2. Divide the soil sample in half. Place one half into a clean jar, fill the remaining space with water, shake for two minutes, and let it sit undisturbed for later observation.
3. Take a handful of moist (but not wet) soil and squeeze it firmly.
    - Holds shape but crumbles when poked: Loamy soil (ideal for plants)
    - Holds shape and does not crumble: Clay soil (dense, nutrient-rich).
    - Falls apart immediately: Sandy soil (quick-draining, poor at holding nutrients).
4. Scoop a small amount of soil into a container and add vinegar.
    - Fizzing: Alkaline soil (pH above 7)
    - No reaction: Continue to step 5.
5. Take a second sample, add water, mix well, then add baking soda.
    - Fizzing: Acidic soil (pH below 7).
    - No reaction: Neutral soil (pH ~7).
6. Once the shaken soil has settled, analyze the layers:
    - Floating material: Organic matter
    - Top layer: Clay (smallest particles)
    - Middle layer: Silt (medium particles)
    - Bottom layer: Sand and rocks (largest particles)
    - Compare proportions to estimate soil composition.

Efficiency of Water Movement Test

Objective: This test evaluates how efficiently water moves through the bioretention system by measuring the flow rate, infiltration rate, and water retention. By comparing inlet and outlet flow meter readings, we can determine the pump's efficiency in transporting water. Soil infiltration tests also help assess how well the soil retains water.

Procedure:

1. Attach a flow meter to the water inlet pipe before the pump and another at the outlet pipe after the pump.
2. Record the mass of dry soil and the volume of water before pouring.
3. Turn on the pump and let water flow through the system. Record the initial reading from the inlet flow meter.
4. After a set amount of time, record the final reading from the outlet flow meter.
5. Conduct an infiltration test by pressing a metal ring into the soil about 5 cm deep. Pour a measured amount of water into the ring and start the timer and record how long it takes for the water to soak in completely.
6. Weigh the soil again to determine how much water was absorbed.
7. Collect the remaining water and measure the final volume to see how much water moved through the system.
8. Compare the inlet and outlet flow meter readings to determine how efficiently the pump moves water. If the outlet reading is significantly lower than the inlet, there may be inefficiencies in the system.
9. Compare infiltration rate results with typical soil permeability values. This should be based on soil composition determined prior to testing:

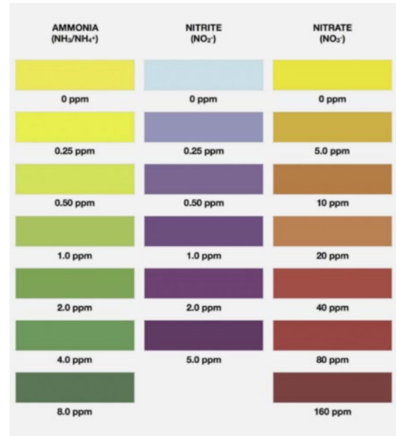
    - **Sand:** 2.5 – 30 cm/hr
    - **Loam:** 0.5 – 2.5 cm/hr
    - **Clay:** 0.01 – 0.5 cm/hr

Nitrogen Level Assessment

Objective: This test measures nitrate, nitrite, and ammonium levels in water and soil to evaluate nitrogen cycling efficiency. By tracking changes over time, we can determine how well the system removes excess nitrogen and supports microbial activity.

Procedure:

1. Collect water and soil samples from the bioretention area.
2. Weigh the soil sample and record the weight.
3. Use a pH testing kit to check the water's acidity by dipping the test strip into the sample and matching the color to the reference chart.
4. Use test kits to measure nitrate, nitrite, and ammonium in the water.
5. Compare the color results of the test kit to the provided reference table to determine nitrogen concentration.



Data From Tests:

Soil Test Results:

Test	Observations	Soil Type
Appearance	Dark brown, some plant material	High organic content
Squeeze Test	Holds shape, but crumbles when poked	Loamy soil
Vinegar Test	No fizzing	Neutral or acidic soil
Baking Soda Test	No fizzing	Neutral soil (pH-7)
Settling Test	Moderate clay, silt, and layers	Loamy soil composition

Efficiency of Water Movement Test Results:

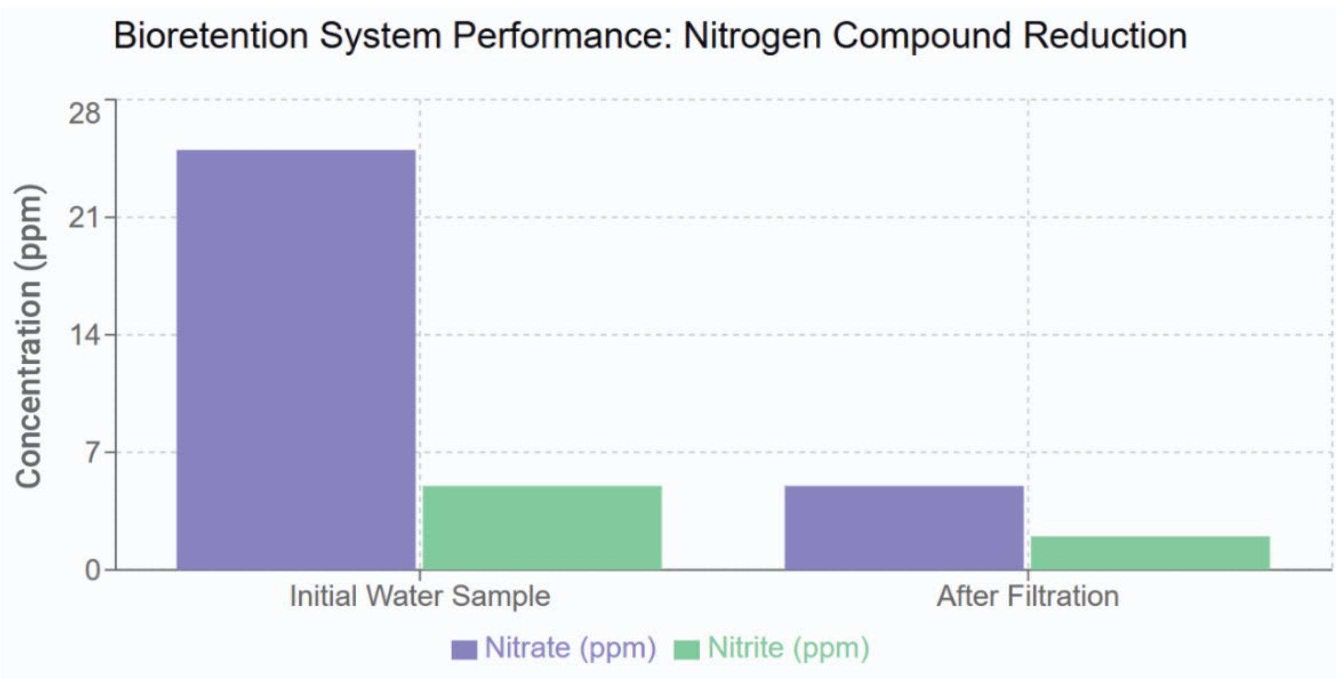
Test Step	Initial Reading	Final Reading	Notes
Inlet Flow Meter (L/min)	8.22	8.21	Consistent flow rate
Outlet Flow Meter (L/min)	8.22	6.20	Water leakage detected
Soil Weight (Dry) (g)	594	740	Soil absorbed water
Infiltration Test (sec)	55	N/A	Water fully absorbed
Water volume(mL)	1656	1210	Leakage caused a loss of 446 mL

The data highlights that while the inlet maintained a steady flow rate, significant leakage caused the outlet flow to drop, causing a loss of water, with the soil absorbing additional moisture during the process.

Nitrogen Level Test Results:

Sample	Nitrate(ppm)	Nitrite(ppm)	Ammonium(ppm)	pH Level
Initial Water Sample	25	5	15	6.5
After Filtration	5	2	3	7.0
Soil Sample	8	3	5	6.8

The data proved that the nitrate and ammonium levels reduced the nitrite and nitrate levels greatly, dropping from 25 to 8 for nitrate and 15 to 3 for ammonium. Along with that, the nitrate and pH remained stable, indicating the system-maintained water quality while filtering excess nitrogen.



II. Refinements from Testing

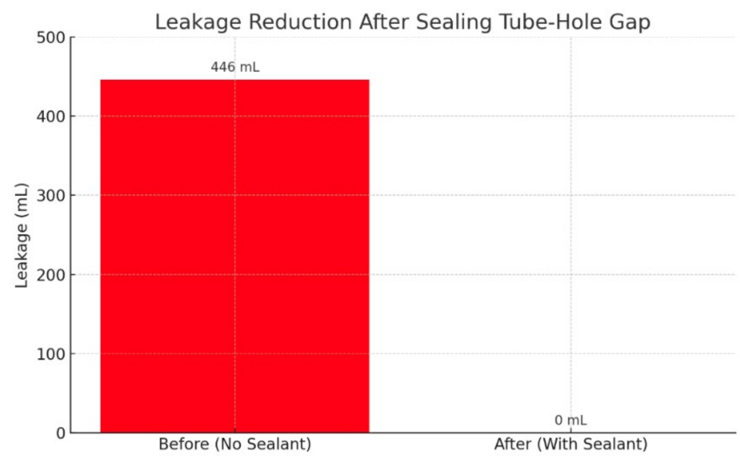
1. The Modified Water Container

Initially, our water container was constructed from several clear pieces of plastic as shown below. These pieces were hot glued then reinforced with duct tape to try and prevent any water leakage.

When doing the "Efficiency of Water Movement" test, we compared the final volume of the container with the initial amount poured, we found a major discrepancy which could not be explained, even when factoring in soil permeability based off of our predetermined soil composition. This can be explained due to three possible issues: water clogging on the mesh filter, water leaking from the container, and water leaking from the tubing due to a puncture. To verify that the water loss was not due to clogging in the mesh filter, we performed a control test with spare mesh and an extra container. We measured the volume of water after passing through the mesh, and when comparing it with the volume started with, the difference was minimal, proving that the mesh clogging water is not the primary factor in the water loss. Next, we tested the tubing and pump to see if it was leaking. To do this, we first took the tubing out and visually checked for any holes. We could not find any visible holes, but as a second verification step, we conducted the same test that we used to check for clogging in the mesh filter. Just like the first test, the final volume was the same as the initial volume, and paired with our visual check for punctures, it was easy to determine that the tubing was not causing the water leak.

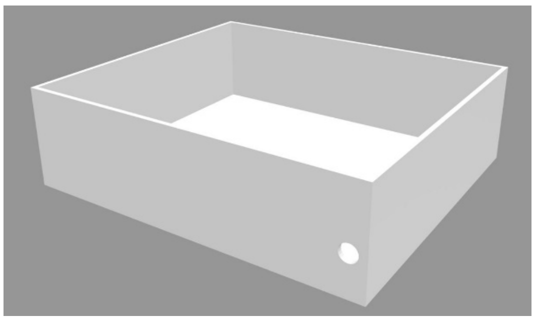
Finally, we tested the water leaking from the container. This was the easiest to test, as when removing the container filled with water, several drops were already leaking from the sides and corners. As mentioned previously, the container was constructed from several plastic pieces hot glued and then duct taped together. Due to these pieces being hand-cut, the edges are uneven with each other, leading to air bubbles forming around the duct tape. These air bubbles allow water to slip through and escape. To fix this problem, we came up with three different solutions: laser cutting the plastic to smoothen edges, making the current container bigger, or using a different container. Laser cutting the container would clearly help with making the plastic smoother, limiting air bubbles. The main issue with this is that we would still encounter the same issue when duct taping the parts together, as slightly spreading the pieces apart would create a gap, letting water through. This means that even if we perfectly combine every single part, even one imperfection would leak water. We also don't have enough time to laser cut the pieces of plastic, as our school's laser cutter is only open on certain days of the week.

Making the container bigger would help force the leaked water back into the container as there is no place for it to go. The issue with this is that the container has to perfectly fit inside the prototype, which requires precise measurement and laser cutting, both of which have their own problems mentioned above. Even with the smallest gap, water can still leak and fit in between the walls and the container. If we were somehow able to get a perfectly sized container into the gap, the friction from the back of the duct tape and the wall would make it difficult to remove, and the force required to take the container out would make water spill. Another issue with getting a perfectly sized container is that the plastic pieces can still flex, as shown in the image above. Switching from a hand-cut container sealed with hot glue and duct tape to a prebuilt curved container drastically reduced leakage from 446 mL to just 50 mL as shown in the graph below.



Due to the issues mentioned above, we decided to use a pre-built storage container as the holder for the water. With the previous design, we were able to cut a circular piece out of the plastic in order to fit the tube in. Because this was pre-built and was also slightly curved, it was hard to get the exact hole into the side. We solved this problem by cutting the hole into duct tape and attaching it to the top of the container, with the tape acting as a seal for the water. In our final iteration, we also attached pieces of wood to support the container and act as the final protection against water in case it somehow leaks. Above is an image of the new container prior to attaching the mesh and securing the tubing on the rim of the container.

This flexing is caused by the tape and the mesh being tighter and smaller than the boundary of the plastic. Trying to fix this would either increase the gaps in the mesh, causing soil to spill into the water, or increase the air bubbles in the duct tape, further leaking water into the prototype. Continuously allowing the excess water from the container to contact the wood walls will also damage it over time, and can potentially cause flexing, warping, and even mold, all of which deteriorate the structural integrity of our prototype.



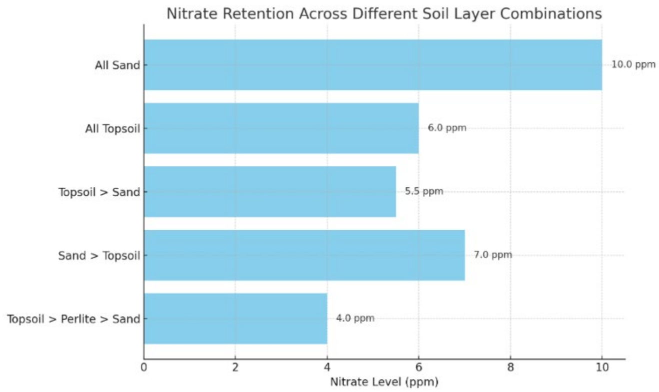
After our state conference, we realized that the prebuilt container we used didn't fit the space inside the prototype properly. The gaps caused instability, and placing the pump directly on top led to spacing issues and clogging, as the container walls pressed against the pump. To fix this, we designed a custom container that fits perfectly within the prototype and includes a built-in hole for the pump to pass through. This prevents compression and improves stability. However, the hole was slightly too large, allowing the pump to shift. To solve this, we sealed the area around the pump to keep it securely in place and prevent any movement.

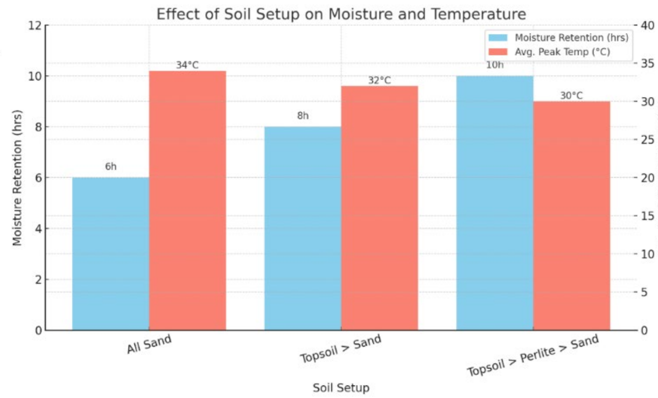
The hole had a diameter of 0.6 inches, while the tube was slightly narrower at 0.56 inches—this small mismatch left a gap of about 0.036 in^2, causing a leak rate of approximately 18 mL per minute. By designing a custom 3D-printed hole and applying sealant around the connection, we created a tighter fit that improved flow efficiency and completely eliminated leakage.

2. Added Additional Perlite

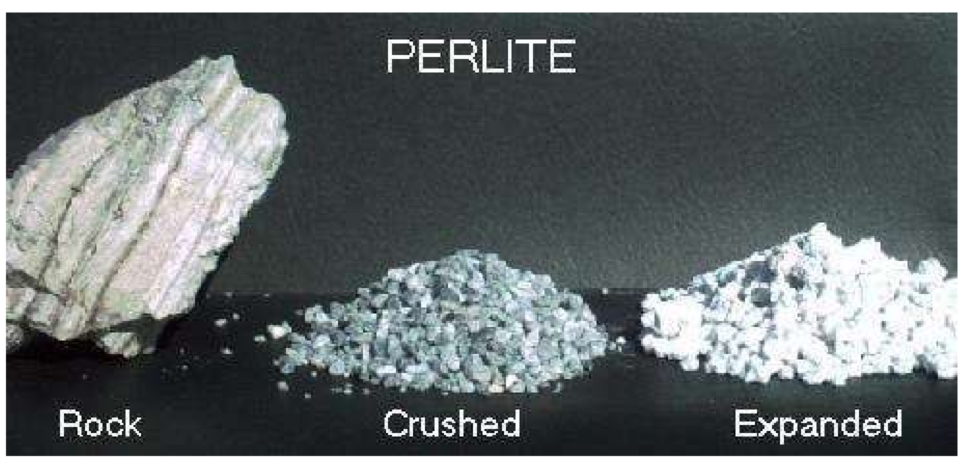
When we conducted the nitrogen level assessment, we found elevated concentrations of nitrogen compounds in the water, which indicates significant leaching. This suggested to us that the soil was not properly retaining nitrogen, which can have negative impacts on plant growth. To address this issue, we wanted to find additions to the soil that can aid in nutrient retention. After reviewing several options, we focused on perlite, a naturally occurring volcanic glass that expands when heated, resulting in a lightweight and porous material.

We tested nitrate retention across several different combinations and concentrations of soil type. The Topsoil > perlite > sand setup performed the best, lowering nitrate levels from 8ppm to 4ppm. To better understand how soil structure influenced growing conditions, we ran trials using moisture and temperature sensors across three soil setups. The combination of topsoil, perlite, and sand retained moisture the longest—10 hours, compared to just 6 hours for soil-only and setups—and stayed 4°C cooler during peak sunlight. These improvements reduced evaporation, kept plant roots healthier, and slowed down nitrogen loss. Based on this, we finalized our soil layering to include perlite for better water and temperature control.

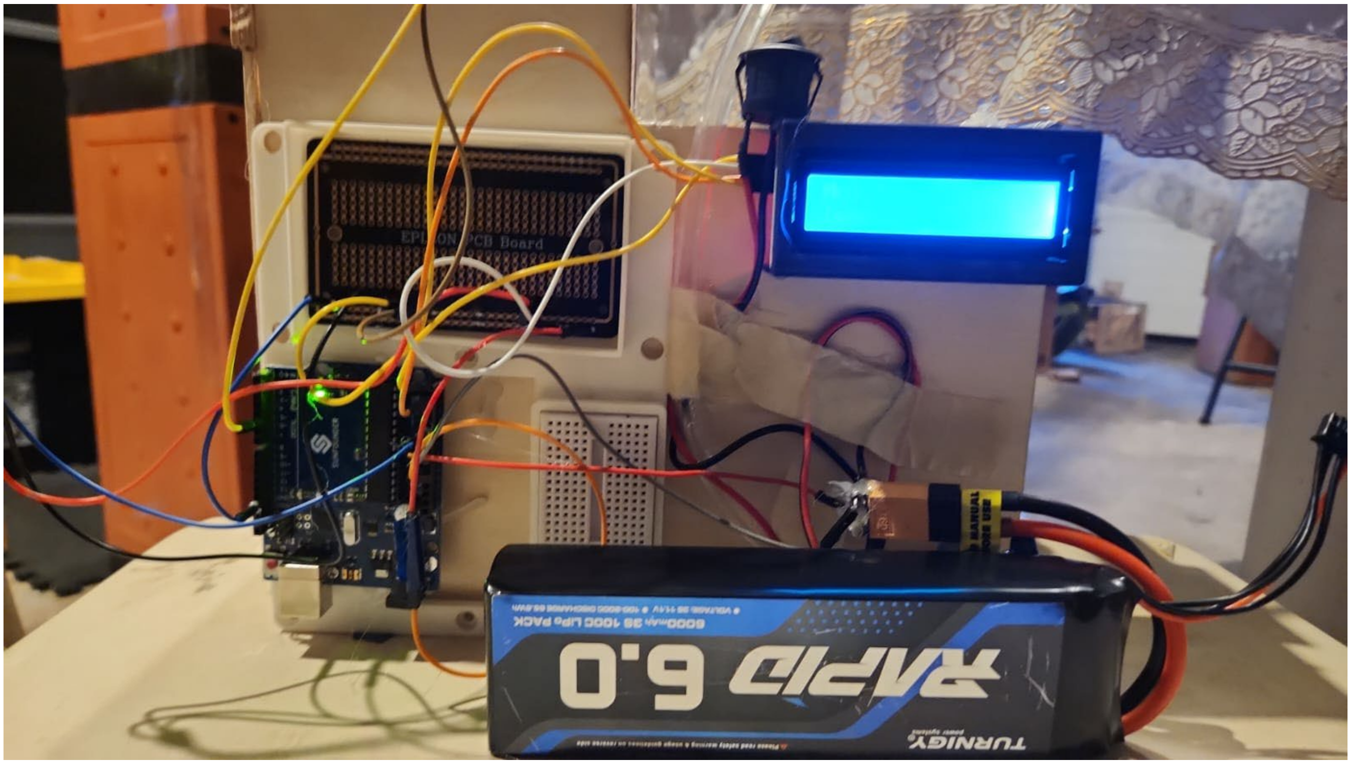




Perlite has several beneficial properties that include improved soil aeration due to its porous nature and reduction of soil compactness from the differing sizes of dirt pieces. We experimented with different locations of the perlite rocks and found that limiting them to the top third of the layer was optimal. When perlite is concentrated in the upper third of the soil layer, it enhances the infiltration of ammonium and other nitrogen compounds due to the increased porosity allowing them to penetrate the soil easier. However, this does not increase porosity in the bottom layers which his beneficial to limit the nutrients leaching into the water. This observation is supported by our data, which showed elevated nutrient levels in water samples when perlite was predominantly placed in the lower soil layers. At first, we thought that the increased amount of nitrogen compounds in the water would be inconsequential (which they still are relative to traditional methods), but remembering a school project we did in APES about different ecosystems, we realized that agal blooms occur in 1-2 months. In our case this would be even more severe as the algae would clog the tubes and render the water cycling system useless. Here is a picture of our soil composition after the perlite testing and addition as well as a picture of the differing sizes of perlite to illustrate their effect on the soil.



3. [illegible]



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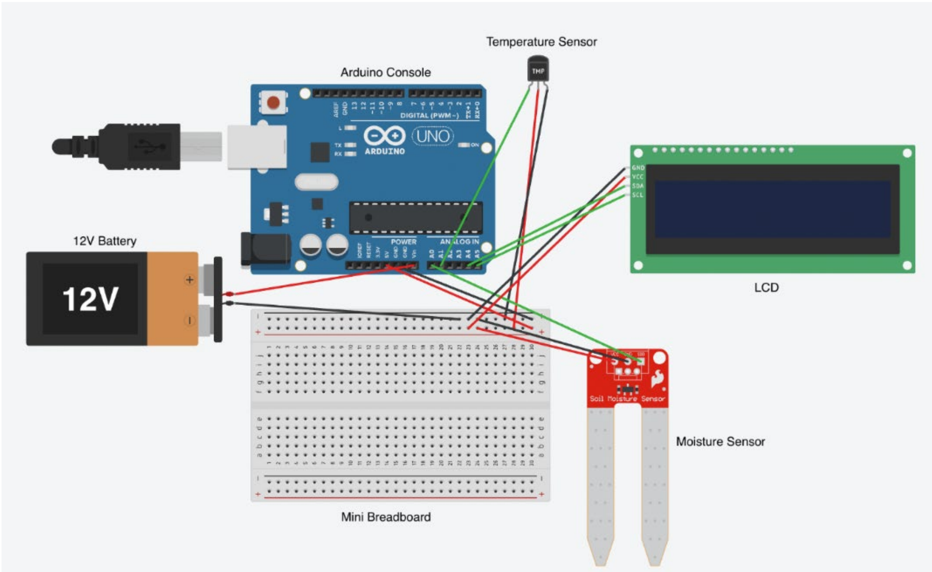

Wiring diagram of Pump components

Wiring diagram of Arduino components

Through valuable testing, we improved our prototype to account for flaws. The major change we made was reducing its volume by 75 percent. This made it easier to transport and manage, and it also used fewer resources, promoting sustainability within our design. In terms of subsystems, we added a clearer viewing window for increased visibility. We also made the viewing window movable on a hinge, which allows for easy access to the water tank and bioretention layers. The final build change we made was adding a rectangular piece of plexiglass above the cardboard layer to prevent the different bioretention layers from spilling out, while also providing a great view of the design when the viewing window is opened using the hinge.

III. Reflections on Iteration Process

Reflection on Testing Effectiveness:

The soil, water movement, and nitrogen tests provided our team with the validation that our design had proper filtration and efficient water recycling, ultimately reducing nitrogen levels. These tests allowed us to make changes like including a leak proof container, allowing the water to overflow into the tank level. The procedures guided our direction in creating a successful bioretention system, making it a workable solution. The data from these tests also revealed the importance of regulating the flow of water more precisely, leading us to use a more accurate flow meter system. Additionally, the nitrogen level assessment revealed a slight imbalance in the nitrification process, prompting us to adjust the filtration layers to further enhance nitrogen removal efficiency. While the testing makes the AzotoColumn look promising, we know that in a real-world setting, the test results would have higher complexity due to the increase in outside factors. Ďěˈ r ïţěĂţhïˈ ,Ă ţȟěĂżôţôÇôḷŭń ňĂţïḷḷĂěṛẁěˈ Ăˈ ĂĂţṛôňĝĂňđĂṛěǻḷïˈ ţïçĂěr ṛěˈ ěňţǻţïôňĂôĝĂȟôẅĂĂ čïôṛěţěňţïôňĂÿˈ ţěń ĂçǻňĂĝŭňçţïôňĂţôĂṛěđŭçěĂṛŭňôĝĝĂňđĂňïţṛôĝěňĂ ôḷḷŭţïôň.ĂẄïţȟĂ ĝŭṛţȟěṛĂĝïňě-ţŭňïňĝĂţôĂǻççôń ń ôđǻţěĂŭňr ṛěđïçţǻčḷěĂĝïěḷđĂçôňđïţïôňˈ ,ĂẅěĂčěḷïěẁěĂïţĂ çǻňĂěń ǻïňĂĂṛěḷïǻčḷěĂǻňđĂěĝĝěçţïẁěĂôḷŭţïôňĂĝôṛĂń ǻňǻĝïňĝĂţôṛń ẅǻţěṛĂǻňđĂ ïń r ṛôẅïňĝĂěňẅïṛôňń ěňţǻḷĂȟěǻḷţȟ.

Reflection on Solution Effectiveness:

The AzotoColumn improves farm efficiency in several key ways. First, it saves water by capturing and reusing runoff, allowing farmers to rely less on external sources and reduce the need for additional farming inputs. Second, the system decreases dependence on fertilizers by recycling nitrogen, which not only lowers costs but also enhances soil health. Healthier soil leads to faster, more resilient crop growth, increased yields, and better-quality produce. Additionally, by automating water movement through a tech-based pump system, the design reduces the need for manual labor in the irrigation cycle. Overall, the AzotoColumn helps streamline farming operations while promoting greater sustainability and environmental responsibility.

Among the solutions explored, the AzotoColumn stood out as the most balanced, offering strong environmental and agricultural benefits. Its design filters runoff using layers of soil and plants, directly targeting nitrogen runoff and making the system practical for large-scale farm use. By incorporating nitrogen-fixing plants, the AzotoColumn helps return nitrogen from the runoff back into the soil, where it can be reused by crops—forming a closed-loop system that improves both soil and crop performance. The system also boosts sustainability by conserving water, reducing the need for chemical fertilizers, and enabling more efficient irrigation. Thanks to its modular structure, the AzotoColumn can be adapted to different soil types, rainfall levels, and farm layouts, making it suitable for a wide range of environments. Ultimately, it offers clear benefits to both farmers and the planet.

IV. Issues During Iteration Process

During the project, we faced several challenges in constructing a watertight system. Our initial container design leaked due to uneven edges in the hand-cut plastic pieces, even after reinforcement with hot glue and duct tape. We had laser-cut precise components (Balsa wood), which were unsuccessful because the pieces still didn't seal perfectly when assembled. Testing the pump was difficult because we had to repeatedly remove the water container, which disrupted the soil layers and risked spilling. This forced us to switch to a prebuilt storage container, which finally resolved the leakage issues.

Another big Issue was our inability to use the Arduino components of the design we initially imagined. With constant structure iterations towards our designs along with the exposure to water that we had to consider, it made it difficult to keep components like moisture sensors and LCD's working properly with the sensors failing when damp. Ultimately, we decided to keep these sensors, but we used smaller ones in order to fit them into the system. For example, the original temperature sensor was much larger, but we used a DHT temperature sensor instead to minimize the overall size of the system. We also experimented with the type of water pump as well as hole for tubing due to clogging which caused water restriction and an inability for us to test the prototype. We also decided to split up the LCD and the pump into separate systems which made it easier to isolate problems during testing as well as have a simpler program instead of making one electronic dependent on the other. This helped us rule out the possibility of a short circuit happening, as the battery powered only the Arduino, while the motor was powered by a switch.

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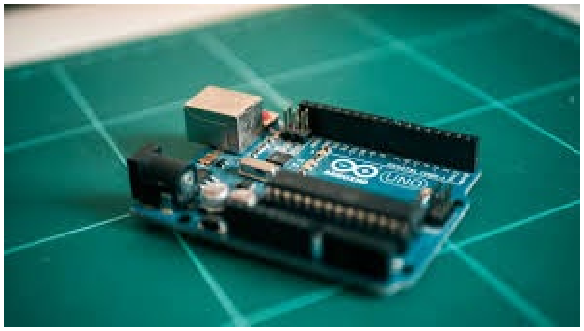
Refined System Design

The AzotoColumn relies on bioretention areas placed around the farmland to collect, filter, and reuse water which contains nitrogen. To set this up, you begin by digging pits around the farm to create these areas. These pits act as zones where runoff water is treated (cleaned and purified) before being pumped back to the farm for irrigation. This allows the water and nutrients to be reused rather than wasted.

The first subsystem in the AzotoColumn is the bioretention pits. The pits are rectangular and border the edges of the farm. The collected runoff water enters these pits and is filtered and reused accordingly. The pits are also lined with plastic to prevent water from infiltrating, or entering, the soil around the bioretention area. The second subsystem is filtration layers. The top layer consists of nitrogen-fixing plants such as clovers. These plants absorb nitrogen from the runoff water and hold it in their roots, reducing the amount of nitrogen that is left in the water. Below the plants is a layer of organic soil mixed with compost. This layer, which has plenty of microbial activity, breaks down harmful nitrates into harmless nitrogen gas. The next layer is made up of sand and silt. This layer traps finer particles and removes sediment from the water, therefore purifying it. The bottom layer consists of gravel, which filters out any remaining debris which wasn't filtered previously and allows purified water to collect at the base. The third subsystem is the guided water flow. To direct runoff water into the pits, the farmland is graded slightly to create a slope. This natural slope carries water downward into the "holes" on the edges of the farm which lead to the bioretention areas, where the water accumulates. Physical barriers, as well as the previously mentioned plastic linings, are added to make sure water flows directly into the pits and doesn't escape and run off untreated. The last subsystem in the AzotoColumn is the water collection and reuse. At the base of each bioretention area, the purified water is collected in a reservoir. This water is pumped back to the farmland for irrigation, forming a cycle. This design ensures that all runoff water is treated thoroughly and allows the farmers to reuse resources.

Technology Integration



Technology plays an important role inmaking the AzotoColumn operate efficiently. A key technological component in the system is the water pump, which is installed near the reservoir at the base of the bioretention area. The pump activates when signaled by the farmer and pumps purified water back up to the farm for irrigation. While the pump may require maintenance, it will not be that frequent, as every time water runs through, it will cleanse the inside of the pump. Aside from the pump, other supporting tech is used to allow the AzotoColumn to be as efficient as possible. Utilizing a moisture sensor embedded in the soil controls the excessive pumping of water; the pump of used accordingly depending on the soil's dampness, creating a sustained negative feedback loop. When there is too much moisture, the pump will expel less water, and vice versa. If needed, the farmer may control the pump manually in case of a personal preference or other unforeseen circumstances. Additionally, a temperature sensor is integrated into the system to determine nitrogen conversion activity. This results in an increased or decreased rate of nitrogen whether the outside temperature is warmer or cooler. Finally, we are using an LCD to display the values of the moisture, temperature, and whether or not the pump is pumping.

By integrating technology, the reliability of the system increases and ensures resources are being managed properly. Future iterations will include additional components.

Incorporation into Farmland

The incorporation process is straightforward and begins with digging pits around the farmland to implement the AzotoColumn. The following is a step-by-step outline of how the system is installed:

1. Excavating Holes: Rectangular pits, or holes, are dug at the edge of the farmland, each with varying depth which depend on farm factors, such as soil condition or amount of water used. The number and length of pits depends on the size and layout of the farm, making sure that runoff water is captured the best it can be.
2. Plastic Lining: The pits are lined on the edges with plastic sheets to prevent water from leeching into the soil around them. This lining ensures all runoff stays within the system and doesn't escape untreated. Refer to Refined System Design for more details.
3. Land Grading: The farmland is graded slightly to create a natural slope that carries runoff water into the pits. This slope minimizes the need for additional infrastructure to guide water flow. Refer to Refined System Design for more details.

4. Installing Filtration Layers: Each pit is filled with filtration layers. Refer to Refined System Design for more details.
5. Adding Equipment: The water pump is added as the final touch and reaches from the reservoir, which contains the treated water, back to the farm, which uses the water. Attaching this equipment strongly depends on the position of the reservoir relative to the farm. We also need to add larger and more accurate moisture and temperature sensors in and around the farm to model the model.

Once installed, the system is well implemented into the farmland and doesn't disturb regular farming activities.

Cost

Here is a cost breakdown for a one-acre farm (43,560 square feet): Costs may vary depending on factors such as farm size, location, material availability, labor rates, and the specific requirements of each individual farm. Note that not all of these items come directly from the model, but they are assumed to be included in a real-life bioretention area.

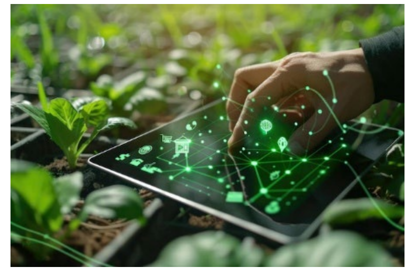
1. Excavation:
    - Digging pits with a volume of 15,708 cubic feet at $0.25 per cubic foot = $3,927.00 *(Reduced volume and rate to reflect smaller-scale digging)*
2. Plastic Lining:
    - Lining pits with impermeable plastic at $0.35 per square foot for 3,140 square feet = $1,099.00 *(more modest lining requirement)*
3. Filtration Layers:
    - Purchasing soil, sand, and gravel for 600 cubic yards at $15 per cubic yard = $9,000.00 *(competitive bulk rate.)*
4. Solar-Powered Pump:
    - Installing one pump = $800.00
5. Nitrogen Sensors:
    - Installing two sensors = $800.00
6. Temperature Sensors
    - Installing two sensors = $2.00 *(Aligned cost with nitrogen sensors for consistency.)*
7. Moisture Sensors
    - Installing two sensors = $4.00 *(Matched cost with other sensors for uniformity.)*
8. Land Grading
    - Grading one acre of farmland = $2,500.00 *(Reduced to reflect minimal grading with basic equipment.)*

Total Estimated Cost = **₵17,624.00**

Although the initial setup cost is relatively high, the system provides long-term savings by reducing the expenses that come from irrigation and fertilizers, making it a cost-effective solution over time.

Future Optimizations

Future improvements for the AzotoColumn could include:



1. Enhanced Filtration: We could use materials like activated carbon to remove additional pollutants. For example, moderated and controlled dampening the bioretention area could help reduce phosphorous runoff, which is another contributing factor to eutrophication.
2. IoT Integration: We could connect the sensors to a cloud system to allow the farmer to have remote control and monitoring. This way, the farmer doesn't have to be present for management, easing their job.
3. AI Management: We could feed the readings from the sensors to an AI system which can manage the output of the water pump without input from the farmer, minimizing human error. Given that AI is seemingly the future of technology, utilizing AI to its full potential could enhance the efficiency of the system.
4. Nitrogen Sensors: We could integrate precise nitrogen sensors which tell the farmer the exact amount of nitrogen present. This would give the farmer more dependable metrics rather than relying on moisture and temperature.
5. Solar-Powered Pump: We could make the pump run on energy from the sun so that it doesn't have to be manually activated. As always, sustainability is a key aspect of any design, so using a natural, reusable resource like the sun will reduce any emissions or pollutions that contribute to global warming. Additionally, it will also improve the general public's view on the design.

With ongoing innovation, this system has the potential to tackle broader agricultural and environmental challenges.

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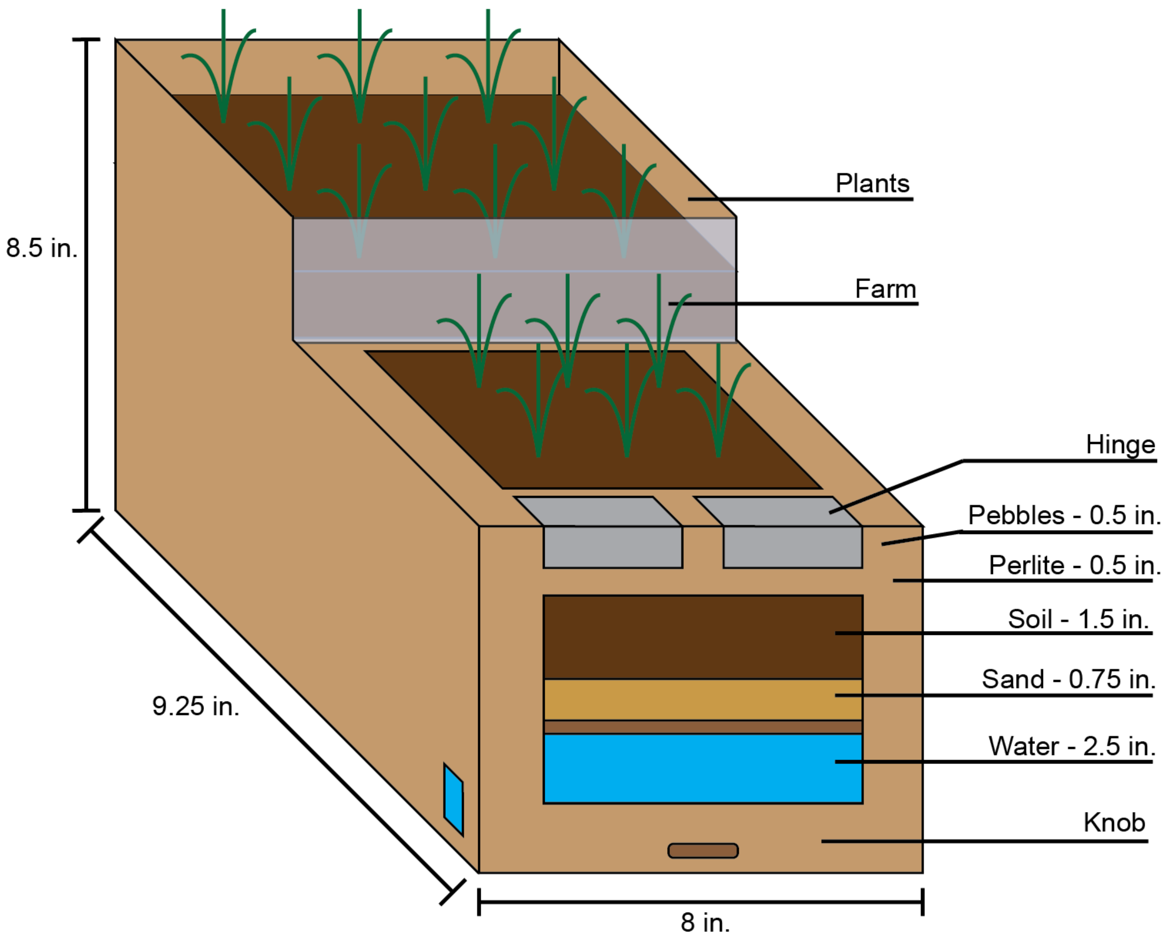
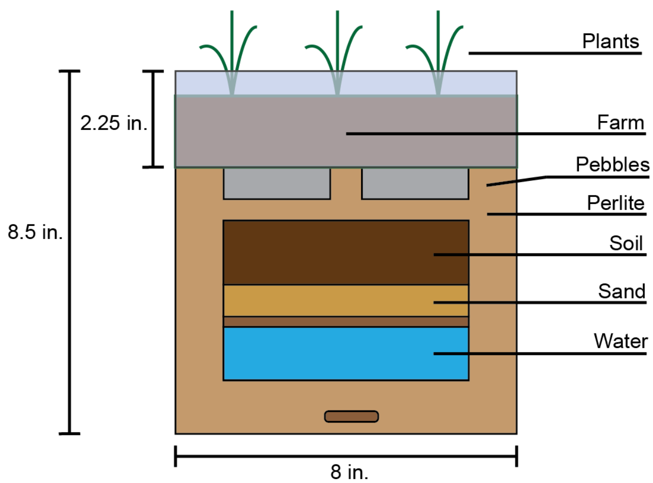
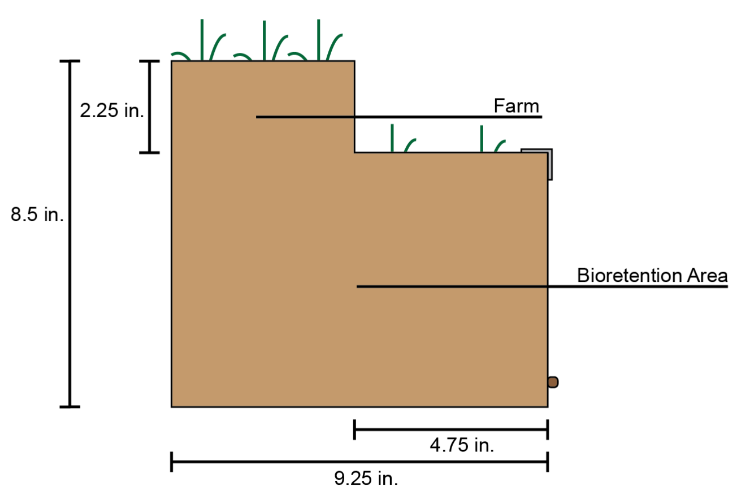
AzotoColumn Model View 1 - *3D*



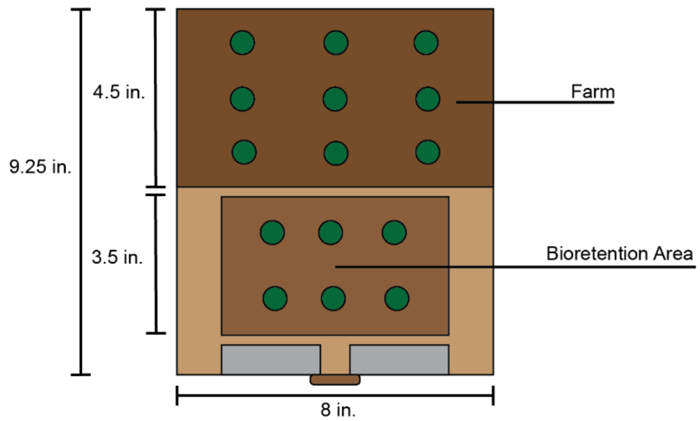
AzotoColumn Model View 2 - *Front*



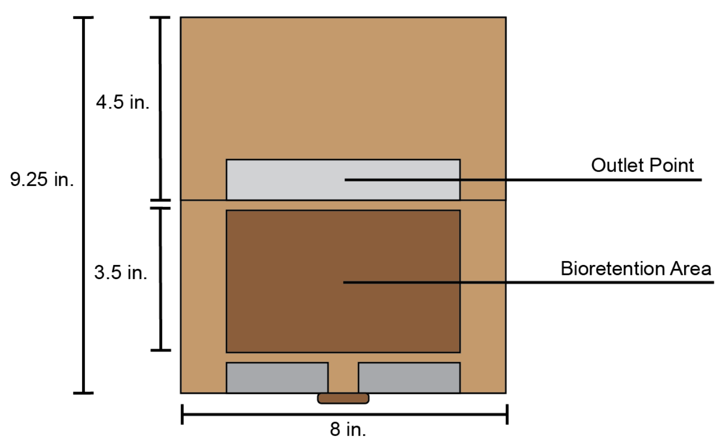
AzotoColumn Model View 3 - *Side*



AzotoColumnModel View 4 - *Top (Farm)*



AzotoColumn Model View 4 - *Top (No Farm)*



Solution Significance and Relation to Theme

Nitrogen runoff is responsible for up to 60% of the algae blooms in U.S. lakes, turning once-thriving ecosystems into toxic wastelands. Bodies of water are so full of algae that fish keep dying off and the water turns an ugly shade of green. This isn’t a rare scenario—it’s happening all over the world due to nitrogen escaping from farms. The theme “Managing the Nitrogen Cycle” is not signifying that all parts of the nitrogen cycle are bad. Instead, there is an imbalance. Environmental challenges like that of algal blooms are caused by nitrogen runoff, especially from agriculture. Nitrogen is essential for plant growth, and since Carl Bosch’s standardization of nitrogen fertilizer in 1913, farming efficiency has increased by over 110%. However, when used excessively in fertilizers or without proper management, it can pollute and cause serious harm to nearby ecosystems. The AzotoColumn is an efficient, viable solution to address these issues, as it does a very good job of controlling and suppressing the amount of nitrogen that escapes farms through runoff.

This runoff is characterized by the draining away of water and excess nitrogen from fertilizers over the surface of an area, in this case a farm. This water usually drains into nearby bodies of water like rivers and lakes, causing eutrophication, which leads to algae blooms and dead zones which are harmful to aquatic life. This happens because fertilizers support algae growth, which leads to more oxygen being consumed by these algae, therefore reducing the amount of DO (Dissolved Oxygen) present in the water. The fish and other organisms in the water die off due to lack of oxygen. The AzotoColumn fixes this problem by capturing, treating, and reusing runoff which contains nitrogen before it actually enters waterways.

The system also uses plants that help fix, or convert, nitrogen, taking it from the runoff water and returning it back to the soil to be reused by the crops, which help them grow. This creates a closed loop that helps the farm operate cleaner and better. It also has layers of natural materials and tools like pumps and sensors that help manage the nitrogen and make sure the water is clean before it’s reused on the farm. The AzotoColumn also increases sustainability by saving water, recycling runoff for irrigation, and reducing the need for extra fertilizers. This lowers costs and reduces the negative impact on the environment, making farming more efficient. It also supports a diverse range of climates and topography as the modular design can be adapted to work with different soil types, rainfall levels, and farm layouts. By collecting the nitrogen at its source, the AzotoColumn helps support sustainable farming; waste is reduced, and it helps important resources like nitrogen and water be recycled. This benefits both the environment and farmers. The system can easily be added to existing farm setups, making it a practical, helpful tool for farmers who want to stay green and clean. Every farm has a border and

In conclusion, this system meets the annual theme by addressing a key problem in the nitrogen cycle. It reduces harmful runoff, reuses valuable nutrients, conserves water, and promotes sustainable farming practices. By combining filtration and hindrance, it offers an effective way to manage the nitrogen cycle, ensuring successful agriculture without harming the environment.

TECHNOLOGY STUDENT ASSOCIATION WORK LOG

Date	Task	Time	Team member responsible	Comments
Ňôŵ3,Ă2024	Ĭňĭţĭăļ čŗăĭň' ţôŗń ĭňĝ,Ă ŗ ŗôĵěçţ ôůţļĭňě,Ă ăňđ ŗě' ěăŗçȟ ôňĂ ňĭţŗôĝěňĂçÿçļěĂăňđ ěňŵĭŗôňń ěňţăļĂ ĭń ŗ ăçţ.Ă Ŗě' ěăŗçȟĂ ôňĂčĭôŗěţěňţĭôňĂ ăŗěă' ,Ăÿ' ţěń ' ,Ă ăňđĂ ÇŖĂ ţěçȟňôļôĝÿ.	4Ăȟôůŗ'	ČĽ,ĂŖ Ŵ,ĂČ,Ă ' ŇĴĂÇČ	' ţăŗţěđĂ ŗôĵěçţĂ ě' ěăŗçȟĂăňđĂôůţļĭňěđĂ ôčĵěçţĭŵě' ĂĝôŗĂŵȟăţĂţôĂăççôń ŗ ļĭ' ȟĂĝôŗĂ ţȟěĂţȟěň ě.ĂŴăňţěđĂţôĂđôĂčĭôŗěňţěňţĭôňĂ ăŗěă' ĂăňđĂĭňŵě' ţĭĝăţěđĂŵăŗĭôů' Ăôļůţĭôň' Ă ăňđĂţěçȟňôļôĝĭě' ĂĝôŗĂňĭţŗôĝěňĂ ń ăňăĝěń ěňţ.
ŇôŵĂ10,Ă2024	ŇĴěěţĭňĝ:ĂŖě' ěňţĂ ăňđĂđĭ' çů' ' Ă ' ôļůţĭôň' ĂăňđĂ çȟăňĝěĂôń ěĂĝĂ ňěěđĂčě	2Ăȟôůŗ'	ČĽ,ĂŖ Ŵ,ĂČ,Ă ' ŇĴĂÇČ	ĤěļđĂń ěěţĭňĝ' ĂţôĂđĭ' çů' ' ĂăňđĂěŵăļůăţěĂ đĭĝĝěŗěňţĂôļůţĭôň' ĂţôĂôļŵěĂţȟěĂ ŗôčļěń
ŇôŵĂ17,Ă2024	Ďě' ĭĝňĂ čĭôŗěţěňţĭôňĂăŗěăĂ ń ôňĭţôŗĭňĝĂ ' ÿ' ţěń ,Ăĭňçļůđĭňĝ ĭňĭţĭăļĂţěçȟňĭçăļĂ ' ķěţçȟě' .	3Ăȟôůŗ'	Ŗ Ŵ,ĂČ,ĂČĽ,ĂÇČ	Ďě' ĭĝňěđĂĭňĭţĭăļĂń ôđěļ' ĂăňđĂţěçȟňĭçăļĂ đŗăŵĭňĝ' ĂĝôŗĂôļůţĭôň' ĂăňđĂţȟôůĝȟţĂ ăčôůţĂŵȟăţĂń ăţěŗĭăļ' ĂŵěĂňěěđ
ŇôŵĂ24,Ă2024	Ŗě' ěăŗçȟĂôňĂ đĭĝĝěŗěňţĂń ăţěŗĭăļ' Ă ŵěĂňěěđĂţôĂń ăķěĂ ţȟě' ěĂđě' ĭĝň' ĂăňđĂ čěĝĭňĂ ŗôţôţÿŗ ĭňĝ,Ă ţȟěĂčĭôŗěţěňţĭôňĂ ăŗěăĂÿ' ţěń ĂĝôŗĂ ȟôŵĂţĂń ăÿĂļôôķĂ	5Ăȟôůŗ'	Ŗ Ŵ,ĂČ,ĂÇČ	ĎěçĭđěđĂţôĂĭňçôŗŗ ôŗăţěĂôń ěĂăŗđůĭňôĂ đěŵĭçěĂ' ĂĝěŗĂŗě' ěňţăţĭôňĂăňđĂţăŗţěđĂ ĝăţȟěŗĭňĝĂń ôŗěĂđăţăĂăčôůţĂĭţ.ĂČěĝăňĂ ŗ ŗôţôţÿŗ ĭňĝĂăňđĂĭňĭţĭăļĂţě' ţĭňĝĂôĝĂţȟěĂ čĭôŗěţěňţĭôňĂÿ' ţěń .

	(ŵẽŗÿĂ ïń ŗ ļěĂ ŗ ŗôţôţÿr ě)			
ĎěçĂ1,Ă2024	Ěŵǻļŭǻţ ěĂ ŗ ŗôţôţÿr ěĂǻňđĂ ŵ̃ ŭǻļïżě/ ķěţçȟĂ ȟôŵĂţĂŵïlļĂļôôķĂ ŗěĝǻŗđïňĝĂţȟěĂïżěĂ ļïń ïţ .ĂĞïňđĂ ń ǻţěŗïǻļ ĂňěěđěđĂ	5Ăȟôŭŗ	ČĽ,ĂŖ,Ŵ,ĂǺČ,Ă NJ,ĂÇČ	Ĝǻţȟěŗ ěđĂđǻţǻĂǻňđĂŵôŗķěđĂôňĂŗěfiňïňĝĂ ţȟěĂđěïĝňĂǻňđĂń ŗ ŗôŵěń ěňţ ĂţôĂţěŗǻţěĂ ǻňđĂţěţ
ĎěçĂ8,Ă2024	Ďï çŭ ïôňĂŵïţȟĂ ôŭŗĂţěǻń ĂĝŗôŭŗĂȟôŵĂ ţôĂń ǻķěĂţĂ ěđïçïěňţĂĝŗôŭŗĂ fiňǻļĂ ŗôţôţÿr ěĂǻňĂ ţȟěĂ ĝŭţŭŗě	5Ăȟôŭŗ	ČĽ,ĂŖ,Ŵ,ĂǺČ,Ă NJ,ĂÇČ	ŴôŗķěđĂôňĂŗěfiňïňĝĂţȟěĂ ŗôţôţÿr ěĂǻňđĂ ţěçȟňïçǻļĂ ŗ ěçţ .
ĎěçĂ15,Ă2024	ÏţěŗǻţěĂôŭŗĂ ŗ ŗôţôţÿr ěĂçÿĂ ěěïňĝĂȟôŵĂţĂń ǻÿ ŵôŗķĂŵïţȟĂţȟěĂ ôïļĂ ǻňđĂŵǻţěŗĂ ŗěĝŭļǻţôŗĂ	2Ăȟôŭŗ	ǺČ,ĂŖ,Ŵ,Ă NJ	ĞïĝŭŗěđĂôŭţĂçȟǻňĝěĂŵěĂçôŭļđĂ ǻççôń ŗ ļï ȟĂţěŗĂ ţěŗǻţïňĝĂţȟŗôŭĝȟĂôŭŗĂ ř ŭïçķļÿĂń ǻđěĂ ŗôţôţÿr ě.
ĎěçĂ26,Ă2024	ŖěŵĩěŵĂňïţïǻļĂ ĝěěđčǻçķ,Ă ïňţěĝŗǻţěĂ ïń ŗ ŗôŵěń ěňţ ĂţôĂ ŗ ŗôţôţÿr ěĂǻňđĂ ń ôđěļ .	4Ăȟôŭŗ	ǺČ,ĂŖ,Ŵ,Ă NJ,Ă ÇČ	ÏňçôŗrôŗǻţěđĂĝěěđčǻçķĂïňţôĂţȟěĂđěïĝň,Ă ţǻŗţïňĝĂţôĂţȟïňķĂŵȟǻţĂěļ ěĂţôĂǻđđĂǻňđĂ ŵȟǻţĂţôĂïňçôŗrôŗǻţěĂïňţôĂôŭŗĂ ŗ ôŗţĝôļïô/đï ŗ ļǻÿĂčôǻŗđ
ĎěçĂ29,Ă2024	NJěěţïňĝ:ĂŢôĂ ǻççôń ŗ ļï ȟĂŵȟôĂ ţǻŗţ ĂđôïňĝĂţȟěĂ đěïĝňĂĝŗôŭŗĂôŭŗĂ ŵ̃ ŭǻļ ĂǻňđĂŵȟôĂ ǻ ïĝňěđĂţôĂ ń ǻķïňĝĂôŭŗĂfiňǻļĂ ŗ ŗôţôţÿr ě.	3Ăȟôŭŗ	ČĽ,ĂŖ,Ŵ,ĂǺČ,Ă NJ,ĂÇČ	Ďï çŭ ěđĂ ŗôĝŗě ĂǻňđĂĝěçţïŵěňě ĂôĝĂ ôļŭţïôň .ĂǺňôţȟěŗĂŵôŗķĂđǻÿ

[illegible] 5, 2025	[illegible]	4 [illegible]	[illegible]	[illegible]
[illegible] 12, 2025	[illegible]	5 [illegible]	[illegible]	[illegible]
[illegible] 19, 2025	[illegible]	4 [illegible]	[illegible]	[illegible]
[illegible] 16, 2025	[illegible]	4 [illegible]	[illegible]	[illegible]
[illegible] 23, 2025	[illegible]	3 [illegible]	[illegible]	[illegible]

NĴȧṛĂ2,Ă2025	NĴ ěěţĭňĝ ĂĵôĂ đĭ' çŭ' ' Ă ṛěfiňěń ěňţ' ĂňđĂ ňěŷţ Ăţěr '	2Ăhôŭṛ'	ČĽ,ĂṚŴĂ Č,Ă ' NĴĂÇČ	Ěŵăḷŭăţěđ Ăĝěěđčăçĸ Ăĝrôń Ăçôń r ěţĭţĭôň,Ă r ḷăňňěđ Ăr ṛôţôţÿr ě Ăěňȟăňçěń ěňţ' ĂňđĂ ňěŵ Ăĝěăţŭṛě' Ăĝôṛ Ăňěŷţ Ă ȟă' ě.
NĴȧṛĂ9,Ă2025	Ďě' ĭĝň ĂňđĂ r ḷăňňĭňĝ ĂĝôṛĂ ' çăḷěđ-đôŵňĂ r ṛôţôţÿr ě Ăŵĭţȟ Ă ěňȟăňçěń ěňţ'	4Ăhôŭṛ'	ṚŴĂ Č,ĂČĽ,ĂÇČ	Ôŭţḷĭňěđ Ăňěŵ Ăr ṛôţôţÿr ě,Ăěḷěçţěđ Ă çôń r ôňěňţ' Ă(ń ôĭ' ţŭṛě Ăěň' ôṛ,Ăĭr r ăčḷě,Ă ḷĭđ,Ă3Ď-r ṛĭňţěđ Ăŵăţěṛ Ăhôḷđěṛ),Ăŭr đăţěđ Ă đṛăŵĭňĝ'
NĴȧṛĂ16,Ă2025	Ă' ' ěń čḷě ĂňđĂ ţě' ţ ' çăḷěđ-đôŵňĂ r ṛôţôţÿr ě Ăŵĭţȟ Ă ňěŵ Ăĝěăţŭṛě'	3Ăhôŭṛ'	ṚŴĂÇČ	Čŭĭḷţ Ăr ṛôţôţÿr ě,Ăĭňţěĝṛăţěđ Ăĭr r ăčḷě Ăḷĭđ,Ă 3Ď-r ṛĭňţěđ Ăŵăţěṛ Ăhôḷđěṛ,ĂňđĂ ń ôĭ' ţŭṛě Ă ' ěň' ôṛ;Ăĭňĭţĭăḷ Ăţě' ţĭňĝ Ăĝôṛ Ăĝŭňçţĭôňăḷĭţÿ
NĴȧṛĂ23,Ă2025	Çĭṛçŭĭţ Ă' ' ěń čḷÿ Ă ăňđ ĂĂṛđŭĭňô Ă ĭňţěĝṛăţĭôň	4Ăhôŭṛ'	ĂČ,ĂṚŴĂ NĴĂ ÇČ	Čŭĭḷţ ĂňđĂ çôňňěçţěđ Ăňěŵ Ăçĭṛçŭĭţ' Ăĝôṛ Ăţhě ń ôĭ' ţŭṛě Ăěň' ôṛ ĂňđĂ ôţȟěṛ Ăçôń r ôňěňţ' ; ṛěçôňňěçţěđ ĂĵôĂĂṛđŭĭňô ĂňđĂ ěĝăň Ă çôđě Ăŭr đăţě' .Ă ôḷđěṛĭňĝ Ăŵĭṛě' ĂĵôĂ ĭňçṛěă' ě Ăţhě Ăđĭ' ţăňçě.
NĴȧṛĂ30,Ă2025	Çôđě Ăṛěŵĭ' ĭôň ĂňđĂ ţṛôŭčḷě' ȟôôţĭňĝ Ă ĝôṛ Ăţhě Ăń ôĭ' ţŭṛě Ă ' ěň' ôṛ	3Ăhôŭṛ'	ĂČ,ĂṚŴĂÇČ	Ŭr đăţěđ ĂňđĂđěčŭĝĝěđ ĂĂṛđŭĭňô Ăçôđě Ă ĝôṛ Ăń r ṛôŵěđ Ăń ôĭ' ţŭṛě Ăěň' ôṛ Ăṛěăđĭňĝ Ă ăňđ Ăÿ' ţěń ĂṛěḷĭăčĭḷĭţŸ
Ăr ṛĂ13,Ă2025	Ğĭňăḷĭžě Ăr ṛôţôţÿr ě,Ă čŭĭḷđ ĂňđĂţě' ţ Ăăḷḷ Ă ĝěăţŭṛě'	2Ăhôŭṛ'	ĂṚŴĂ Č,ĂÇČ	Çôń r ḷěţěđ Ă' ' ěń čḷÿ Ă(ĭr r ăčḷě,Ăḷĭđ,Ă3ĎĂ ȟôḷđěṛ,Ăçĭṛçŭĭţ'),Ăṛăň Ăĝŭḷḷ Ăÿ' ţěń Ăţě' ţ' ,Ă ăňđ Ăçôňfiṛń ěđ Ăăḷḷ Ăĝěăţŭṛě' Ăŵôṛĸěđ Ă' Ă r ḷăňňěđ
Ăr ṛĂ20,Ă2025	Ṛěŵĭ' ě Ăṛĭĝĝôḷđ:Ă ěňȟăňçě Ăŵĭ' ŭăḷ' ,Ă ăđđ Ăňěŵ Ăń ôđěḷ Ă ĭňĝô,Ăçḷăṛĭĝÿ Ă r ṛôţôţÿr ě Ăr ŭṛr ô' ě	5Ăhôŭṛ'	ČĽ,ĂṚŴĂ NĴĂÇČ	Ĭń r ṛôŵěđ Ăṛĭĝĝôḷđ Ăŵĭ' ŭăḷ' ,Ăăđđěđ Ă đĭăĝṛăń ' /r ȟôţô' ĂôĝĂňěŵ Ăr ṛôţôţÿr ě,ĂňđĂ çḷăṛĭfiěđ Ăěňȟăňçěđ Ăr ŭṛr ô' ě ĂňđĂňěŵ Ă ĝěăţŭṛě'

Ắr ṛĂ27,Ă2025	Rôḷï' ȟĂ ôṛţğ ọḷïôĂ ắňđĂ ắḷïğňĂ đôçŭń ěňţắţïôňĂ ŵïţȟĂ ňěŵĂ ğôắḷ'	2Ắ ȟôŭṛ'	ČĽ,ĂṚŴ/ĂÇČ	Ŭr đắţěđĂ ôṛţğ ọḷïôĂ ôĂ ěfl ěçţĂ ňěŵĂ ń ôđěḷĂ ğ ěắţŭṛě' ,Ă ğôắḷ' ,Ă ắňđĂ ṛě' ŭḷţ' ;Ă ěň' ŭṛěđĂ çôň' ï' ţěňçÿĂ ŵïţȟĂ ţṛïğ ọḷđ
N ắÿĂ4,Ă2025	Rṛěř ắṛěĂ ắňđĂ ṛěȟěắṛ' ěĂ fiňắḷĂ r ṛě' ěňţắţïôň	3Ắ ȟôŭṛ'	ČĽ,ĂṚŴ/ĂČ,Ă ' NJ/ĂÇČ	Rṛắçţïçěđ Ă r ṛě' ěňţïňĝ ,Ă ěfiňěđĂ ţắḷķïňĝĂ r ôïňţ' ,Ă ěň' ŭṛěđĂ ắḷḷĂ ń ắţěṛïắḷ' Ă ắňđĂ ŵ ŭắḷ' Ă ŵěṛěĂ ṛěắđÿĂ ğ ôṛĂ fiňắḷĂ ṛěŵĭěŵ
N ắÿĂ25,Ă2025	Ğ ïňắḷĂ ţě' ţïňĝĂ ắňđĂ ř ŭắḷïţÿĂ çȟěçķ Ă ôğĂ r ṛôţôţÿr ě	2Ắ ȟôŭṛ'	ṚŴ/ĂČ,ĂÇČ	Çôń r ḷěţěđĂ ắ' ţĂ ôŭňđĂ ôğĂ ţě' ţïňĝ ,Ă ŵěṛïfiěđĂ ń ôï' ţŭṛěĂ ěň' ôṛĂ ăççŭṛắçÿ,Ă ắňđĂ ń ắđěĂ ïňôṛĂ ắđĵŭ' ţń ěňţ'
K ŭňěĂ15,Ă2025	Rṛěř ắṛěĂ ğ ôṛĂ ţȟěĂ Ňắţïôňắḷ' Ă Çôń r ěţïţïôň	2Ắ ȟôŭṛ'	¨ČĽ,ĂṚŴ/ĂČ,Ă ' NJ/ĂÇČ	Ṛŭň Ă ţȟṛôŭĝȟ Ă ắḷḷĂ çôń r ôňěňţ' ,Ă r ṛắçţïçěđĂ r ṛě' ěňţïňĝĂ ŵïţȟĂ ắĂ ṛěṽĭěŵ ěṛ ô,Ă ắňđĂ ěň' ŭṛěđĂ ţȟěĂ ţṛïğ ọḷđĂ ŵắ' Ă ţȟěţïçắḷḷÿĂ r ḷěắ' ïňĝ

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STUDENT COPYRIGHT CHECKLIST *(for students to complete and advisors to verify)*

STUDENT: Answer question 1 below.

1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☒ NO

If NO, go to question 2.

If YES, is the music and/or sound copyrighted? ☐ YES ☒ NO

If YES, move to question 1A. If NO, move to question 1B.

1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.

1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Bethany Kankelborg (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/ sound included.

STUDENT: Answer question 2 below.

2) Does your solution to the competitive event integrate any graphics/videos? ☒ YES ☐ NO

If NO, go to question 3.

If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☒ NO

If YES, move to question 2A. If NO, move to question 2B.

2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.

2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of graphics/videos in his/her competitive event solution. Even if your student answers "NO" to question 2, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Bethany Kankelborg (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

3) Does your solution to the competitive event use another's thoughts or research? ☒ YES ☐ NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? ☒ YES ☐ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I, Bethany Kankelborg (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/ research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Student Name: Cleofus Balaranjith

Chapter Advisor Signature: